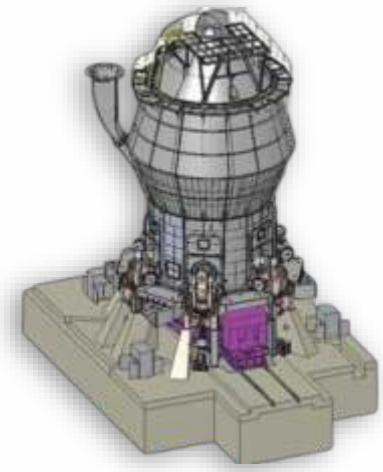
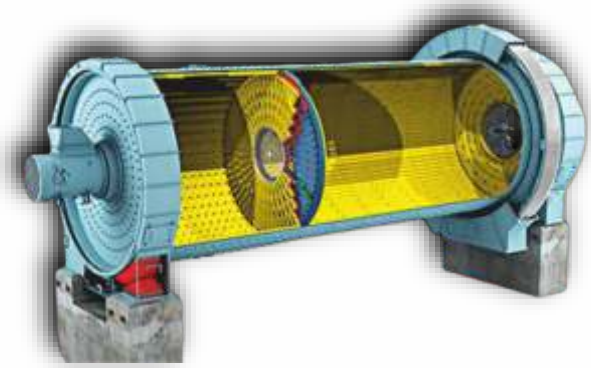
Gujarat Nippon Group

'Service with Quality'





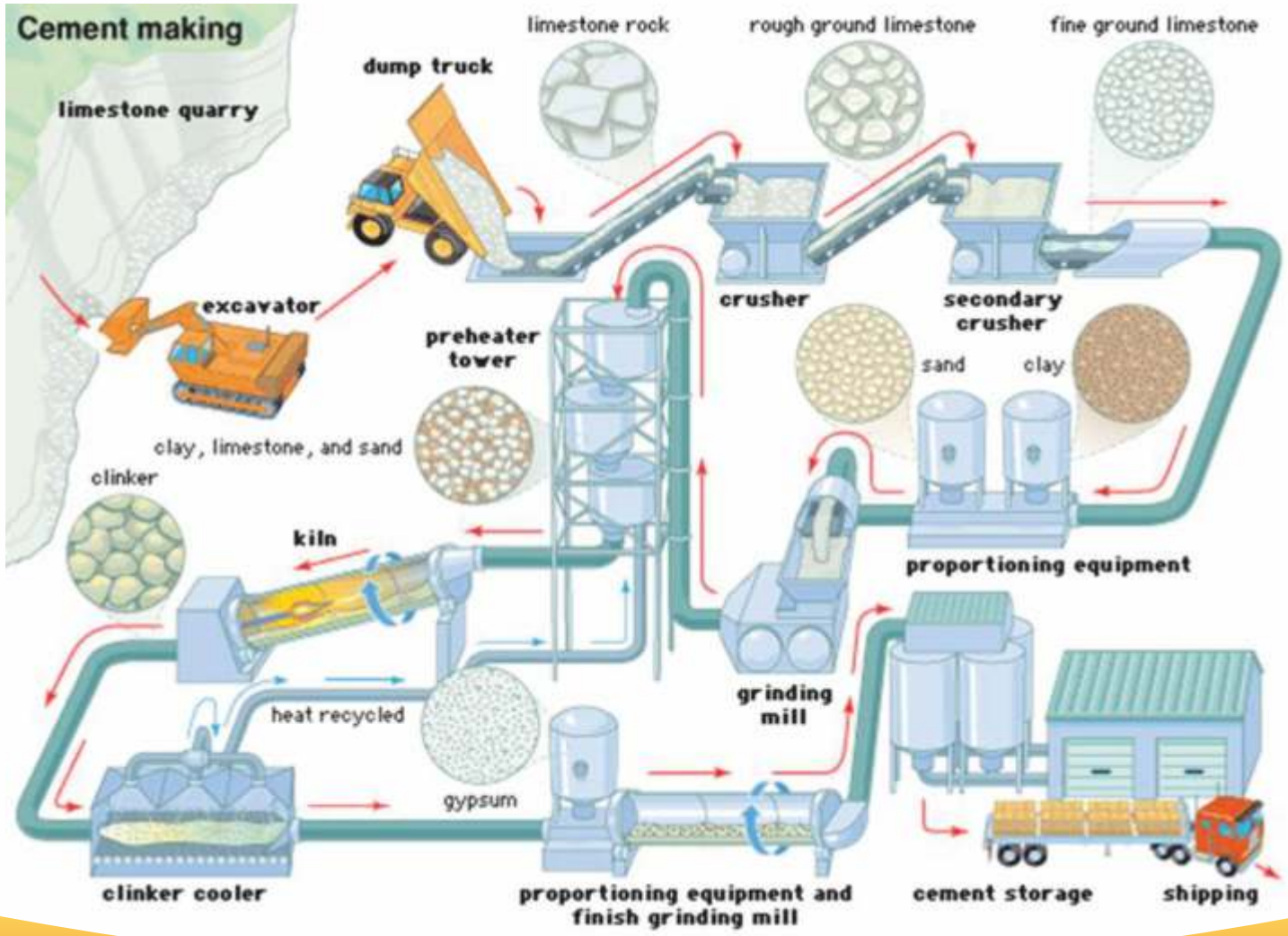
Cement Plant Consumable & Spare Part Supplier

INTRODUCTION

M/s Gujarat Nippon International, source & supply spares, engineering equipment, consumable and services for cement industries. We also provide repairs and return services for capital equipment's .

GNIPL team have a group of experienced engineer's having expertise in their respective fields. We are able to source and supply quality product for **Cement plant .**

CEMENT INDUSTRY GENERAL LAYOUT



OUR SUPPLY RANGE

BUCKET ELEVATOR

Buckets, Belts, Chain, Drive and Driven Drums & Additional spares

BRICKS

Bricks for **cement** rot ary kiln operating under severe conditions with high thermal loads some of the bricks which we can supply are

Fire Clay Bricks, Fused Magnesia Bricks, Magnesia Chrome Bricks, High Alumina Anchor Bricks, Alumina Clay Fire Bricks, Castable Refractory.

BAG HOUSE & FILTER BAGS

BULK LOADING SPOUT

LINERS

Liners inside a Ball Mill

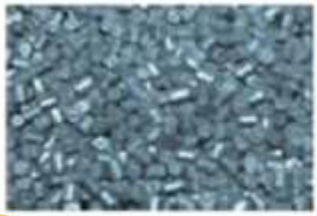
Various Shapes of Cast Steel liners

Cast Steel Diaphragm

Grinding Media

Hi Chrome Cast Steel Media

Cast Steel Roller Type Media



FILTERS

AIR SLIDE CLOTH

OTHER SPARES

01 CRUSHER, STACKER, RECLAIMER

• Scrappers

• Rail Clamps

• Wheels

• Bearing Housing

• Foundation Bolts

• Spindle

• Pull Rod & Tie Rod

• Shafts

• Fish Plate

• Blow Bar

• Digging Spikes

02 RAW MILL SPARES

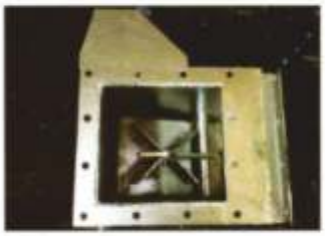
• Impellers

• Guide Vanes

• Lower & Upper Discharge Rings

• Rotor Shaft

03 PRE-HEATER & GRATE COOLER SPARES



• Flap Box Assembly

• Dip Tube (For Cyclone)

• Cone

08 CEMENT MILL

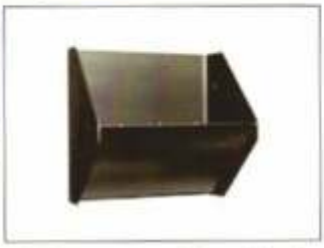
• Air/Fluid Slide Divertor

• Rotor Shaft

• Slide Gate

• Impellers/ Separator Fan

04 CONVEYORS & FEEDERS



- Elevator Buckets

- Deep Pan Conveyor Buckets

- Screw Conveyors

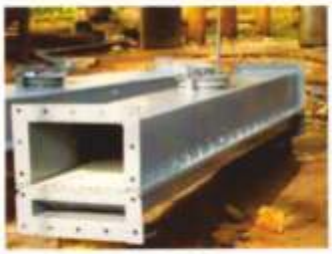
- Paddle Conveyors

- Apron Feeder

05 KILN SPARES

- Kiln Guide Roller
- Retainer Ring
- Burner
- Kiln Outlet Lamella Seal/ Inlet Lamella Seal (Leaf Plates)

06 PACKING PLANT



- Air Slides
- Wagon Loader
- Truck Loader

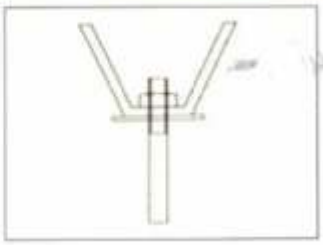
- SS Nozzle

09 FASTNERS

07 REFRACTORY ACCESSORIES (IN MATERIAL STAINLESS STEEL GRADE 304, 309, 310, 316)

• U Anchor

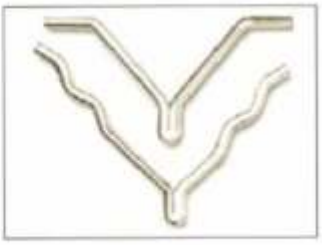
• V Anchor



• Stud Anchor

• Spiral Anchor

• Y Anchor



• Zig Zag Anchors

10 POWER TRANSMISSION EQUIPMENTS

• Flanges & Rings

• Shafts

• Forged Crane Wheels

• Roller Assembly

• All type of Pulleys with and without Rubber Lagging

• Sprocket

11 OTHERS

• Guide Bracket

• I Beam

• Protection Cap

• Partition Plate for DPC Bucket

• Double pendulum flap valve

• SS Nozzle Cap

• Trolley 70T

EQUIPMENTS & SPARES

CIRCULAR MOTION VIBRATING SCREENS

JAW CRUSHER TYPE OM

To crush large feed sizes with maximum reduction ratio

Linear Motion Vibrating Screens

For handling medium to large bulk solid flows

Magnetic Separator

Surface Drill Rig

Compressor & Parts

LIFE ENHANCEMENT SERVICES

COARSE FEED VALVE/ROTARY AIR LOCK

TOTAL REFURBISHMENT OR NEW MANUFACTURE

AREA OF CONCERN

The body & rotor wears out due to heavy handling of material & it's abrasiveness. The gap between rotor blades & body increases making it difficult to control feed rate and smooth operation of rotor.

SOLUTION

Refurbish the body and rotor with added features like use of 'Hardox' or 'Enduraplates' or 'Cast-fused' liners for enhanced resistance to severe abrasion. The performance life is enhanced considerably.

KEY PROCEDURE

- Removal of worn-out parts
- Fabrication of new parts
- Machining of rotor & body
- Fitting of new liners & assembly
- Balancing & finishing

ROLLER PRESS – ROLLERS

TOTAL REFURBISHMENT, IN-SITU PROFILING OR IN-SITU REBUILDING, SHAFT REPAIR

AREA OF CONCERN

a) Rollers getting wear out due to heavy handling of material & it's abrasiveness, under very high pressure. The gap between rollers increases making it difficult to control required size.
b) Chunks of base material coming out of roller, disturbing it's balance and grinding efficiency.

KEY PROCEDURE

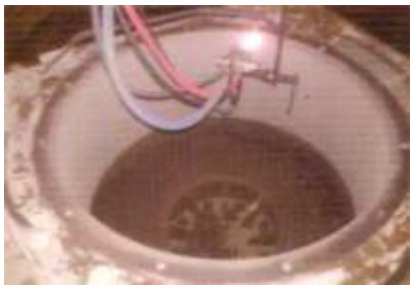
- Removal of old hard deposit & levelling of surface
- NDT of roller & shaft
- Repair of cracks if any and filling of craters locally
- Stress relieving
- Weld build-up & hard deposit
- Profiling & finishing
- Inspection

SOLUTION

Refurbish roller by removing old deposit, rebuilding it by automated WOL process and profiling – all in controlled welding conditions and temperatures.
The performance life is enhanced considerably and we ensure that Chunks won't come out during operation.

ROLLER PRESS BEARING HOUSINGS

TOTAL REFURBISHMENT OF ID I.E BEARING SEAT



AREA OF CONCERN

The ID of the bearing housing wears out due to relative motion between bearing and housing. This results in free rotation of bearing and may cause loss of lakhs of rupees due to seizure or breakage of bearing. The cost of bearing, cost of bearing change and loss of productivity are main concern.

SOLUTION

Refurbish the body at ID and carry out machining.

CRITICAL

Tolerance and co-centricity of repaired area.

KEY PROCEDURE

- Undercut
- Build-up of worn out surface
- NDT of roller & shaft
- Repair of cracks if any
- Stress relieving
- Finished machining
- Inspection

KILN – TYPE CRACK REPAIRS

IN-SITU REPAIR OF ALL CRACKS DETECTED IN KILN TYPES

AREA OF CONCERN

The crack may develop rapidly in tyres if continued to work in the kiln. Tyres are casted and not readily available. Lead time to supply may vary from 4 months to a year and mostly imported. Huge down time resulting in production loss as well as cost involved in tyre replacement. Developed CREEP may cause further damage to other parts of Kiln if tyre is allowed to run without repair.

SOLUTION

Detect and repair cracks in tyre, carry out PWHT and run the kiln till new tyre arrives. Ador has performance proven welding consumables and Quick Response Team to undertake this repair work.

KEY PROCEDURE

- NDT of tyre to mark exact location of cracks
- Arresting cracks to prevent further propagation
- Removal of cracks & surface preparation
- Weld build-up
- PWHT & finishing
- Inspection

KILN – SHELL JOINTS BY SAW

CARRY OUT WELDING OF SHELL JOINTS BY AUTOMATED PROCESS THUS REDUCING PRECIOUS SHUT-DOWN CONSIDERABLY AND HUGE SAVING WITH HIGHEST QUALITY WELD JOINTS

AREA OF CONCERN

MMAW of joints may take up to 6 to 8 days depending upon thickness of shells. Thus, warranting more time of shutdown. This is essential activity and can't be avoided. Being MMAW, quality of welding within a joint itself may vary depending upon welder's skill. Uneven fusion, development of stresses and non-uniformity of welding may lead to bad quality.

SOLUTION

Deployment of SAW process cuts down welding time by at least 4 to 5 days! Thus, saving lots of shut-down time. Strictest control over welding parameters ensure uniform weld bead quality, generation of lesser/negligible stresses and almost ZERO defects.

KEY PROCEDURE

- Root run welding & gouging
- NDT of shell & joint
- Repair of cracks/defects if any
- SAW
- Inspection

KILN – JOBS ON TURN-KEY BASIS

FABRICATION & REPLACEMENT OF KILN, SHELLS, COWL, TYRES, GIRTH GEAR CHANGE OR REVERSAL ETC.

AREA OF CONCERN

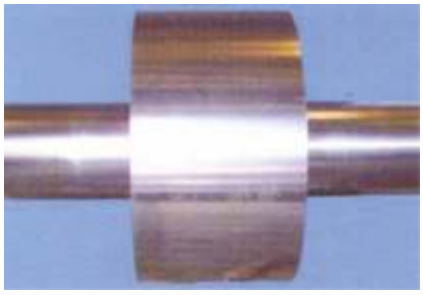
None. ADFL offers turn-key solution with fabrication and services at one stop.

KEY PROCEDURE

- Fabrication of shell, NDT and dispatch
- Mobilization of crew & equipment
- Preparatory work before shutdown
- Runout & ovality measurement, Shell cutting & fitting of new shell
- Alignment and welding
- NDT
- Inspection

KILN-SUPPORT AND THRUST ROLLS

TOTAL REFURBISHMENT WORN OUT ROLLERS OR REPAIR OF CRACKS



AREA OF CONCERN

The support roller body wears out TAPER due to heavy load of kiln. Shaft also develops scratches at seal area. Thus, uneven load on rollers may tend to develop cracks and oil leakage.

SOLUTION

ADFL offers rebuilding of barrel of rollers and repair of shafts, at the same time maintaining surface finish, critical dimensions and stress-free component. Refurbishment id done by welding as well as thermal spray coatings.

KEY PROCEDURE

- Undercut over roller & shaft & levelling of surface
- NDT of roller & shaft
- Repair of cracks if any and if repairable
- Weld build-up & Stress relieving
- Finished machining
- Inspection

BALL MILL – TRUNION HEAD CRACK REPAIR

IN-SITU REPAIR OF ALL CRACKS DETECTED IN BALL MILL

AREA OF CONCERN

The crack may develop rapidly in mill heads if mill is continued to work. The mill Trunion heads are very thick and may be accessible only from out. Changing mill heads warrants availability of mill head which is very remote and also lead time to supply may vary from 4 months to a year. Huge down time resulting in production loss as well as cost involved in replacement. Developed CREEP may cause further cracks in head if mill is allowed to run without repair.

SOLUTION

Detect and repair cracks in mill head, carry out PWHT and run the mill till new mill head arrives. Ador has performance proven welding consumables and Quick Response Team to undertake this repair work.

KEY PROCEDURE

- NDT of Trunion to mark exact location of cracks
- Arresting cracks to prevent further propagation
- Removal of cracks & surface preparation
- Weld build-up
- PWHT & finishing
- Inspection

CRUSHER-ROTORS

TOTAL REFURBISHMENT OR NEW MANUFACTURE

AREA OF CONCERN

Rotor discs & Shaft wears out due to heavy handling of material & it's abrasiveness. The enlarged pin diameter and thinning of discs makes it difficult to operate rotor smoothly. Many times, blow bar holder also breaks making it impossible to operate.

SOLUTION

Refurbishment is carried out by replacement of all discs but retaining used shaft in case of hammer crusher. For Blow Bar Crusher Rotor, total health check-up of rotor is carried out for cracks etc.

KEY PROCEDURE

- Removal of old discs
- Levelling of shaft surface
- NDT of shaft
- Procurement, cutting & machining of new discs
- Fitting and alignment of new discs over used shaft
- Joining of new discs by welding and Stress relieving
- I lard deposit over prone area of discs
- Line boring & finishing
- Inspection

VERTICAL ROLLER MILL – ROLLERS AND HUBS

TOTAL REFURBISHMENT AT OD &/OR ID OR REPAIRS OF CRACK

AREA OF CONCERN

The OD or ID of roller / hub gets wear out due to relative motion of tyre or bearings, loosening tyre or bearing or both making the roller in operational.

SOLUTION

Roller's MOC is usually either Cast Steel or Cast Iron. Rebuilding of surface in case of later is very expensive comparative ot former. Hence, it is essential to know MOC before we quote. Undercut, NDT and finished machining is done.

KEY PEOCEDURE

- Undercut over roller & levelling of suface
- NDT of roller
- Repair of cracks if any and if repairable
- Weld build-up & Stress relieving (PWHT)
- Finished machining
- NDT
- Inspection

VERTICAL ROLLER MILL – TYRES AND TABLE LINERS

REBUILDING IS DONE BY WOLPOCESS USING FLUX CORED WIRES

AREA OF CONCERN

The tyre / shell & table liners wear out due to heavy amount of material handled and it's abrasiveness. The gap between tyre & table liners as well as crushing angle change reducing the output of the VRM thus reducing efficiency and power consumption.

KEY PROCEDURE

- Removal of old hard deposit & levelling of surface
- NDT of rollers / roller segments / tyres and table liners.
- Weld build-up hard deposit and matching profile / angle.
- Finishing
- Inspection

AREA OF CONCERN

Rebuild the worn-out surface by automated, electronically controlled WOL process using flux cored wires. The WOL deposit is far superior than the base material, matallurgically, offering excellent wear resistance. With our FCW, it is possible to rebuild multiple layers for as high as 80mm thickness!

SPECIFIC REQUIREMENT

For faster delivery, please mark all table liners or segments of roller for numbers as well as directions. This will reduce the time required for matching profile, considerably.

VERTICAL ROLLER MILL- CLASSIFIER ROTORS

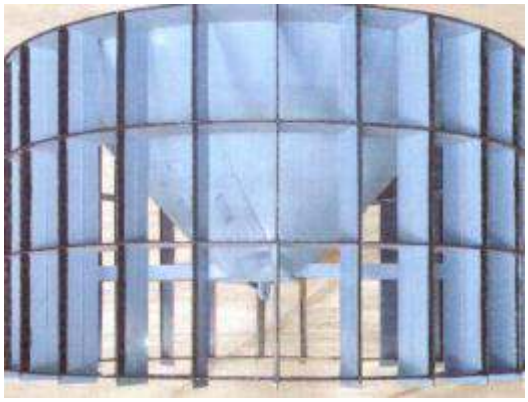
TOTAL REFURBISHMENT OR NEW MANUFACTURE

AREA OF CONCERN

The classifier rotor is subjected to sever, high velocity particle erosion by material being ground in VRM

Solution

Refurbish the rotor by replacing vanes and hard-facing new vanes for added wear resistance. Other options like use of ‘Hardox’ of ‘Enduraplates’ or ‘Cast-fused’ liners for enhanced resistance to severe abrasion are also available. The performance life is enhanced considerably.



KEY PROCEDURE

- Removal of old worn-out vanes & flanges.
- Manufacture of new vanes & flanges with hard facing.
- Assembly, alignment & welding of vanes & flanges.
- Finishing & balancing.
- inspection.